

NICOLET, QC, Canada

WMO: 717230

Lat: 46.226N

Lon: 72.657W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.3	(c) -22.5	(d) -30.4	(e) 0.2	(f) -24.5	(g) -27.6	(h) 0.3	(i) -22.0	(j) 12.2	(k) -1.9	(l) 10.5	(m) -2.7	(n) 2.3	(o) 0	(p) 0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 28.8	(d) 22.4	(e) 27.3	(f) 21.4	(g) 25.9	(h) 20.5	(i) 23.6	(j) 27.3	(k) 22.5	(l) 25.9	(m) 21.5	(n) 24.8	(o) 4.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.1	25.9	21.3	16.0	24.7	20.3	15.0	23.7	70.7	27.2	66.3	26.0	62.8	24.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.3	(b) 8.0	(c) 6.9	(e) -29.4	(f) 31.3	(g) 3.0	(h) 1.3	(i) -31.5	(j) 32.3	(k) -33.3	(l) 33.1	(m) -34.9	(n) 33.8	(o) -37.1	(p) 34.8		
(a) 9.3	(b) 8.0	(c) 6.9	(e) -29.6	(f) 25.3	(g) 2.9	(h) 1.4	(i) -31.7	(j) 26.3	(k) -33.4	(l) 27.1	(m) -35.0	(n) 27.9	(o) -37.1	(p) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-10.6	-9.0	-3.2	5.0	13.0	17.8	20.4	19.3	15.1	8.4	1.6	-5.7
	DBStd	11.81	7.18	6.35	5.63	4.27	3.86	3.45	2.62	2.75	3.82	4.05	4.76	6.10
	HDD10.0	2576	640	533	408	156	14	0	0	0	3	79	254	488
	HDD18.3	4649	898	766	666	399	172	51	9	22	110	307	503	746
	CDD10.0	1148	0	0	0	8	106	235	321	288	157	31	2	0
	CDD18.3	178	0	0	0	0	5	35	72	52	14	0	0	0
	CDH23.3	1099	0	0	0	2	55	234	445	282	80	1	0	0
	CDH26.7	192	0	0	0	0	8	48	85	39	12	0	0	0
Wind	WSAvg	3.1	3.4	3.3	3.3	3.4	3.3	2.8	2.7	2.5	2.7	3.2	3.5	3.4
Precipitation	PrecAvg	1003	73	57	66	79	84	95	104	101	91	95	81	84
	PrecMax	1155	129	120	114	157	151	175	174	213	183	182	130	147
	PrecMin	796	34	26	27	22	35	44	46	47	41	32	28	24
	PrecStd	103	24	20	23	34	29	34	34	33	33	37	23	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	5.5	11.0	20.6	27.6	30.2	31.0	29.6	28.5	22.3	16.0	7.9
		MCWB	5.3	3.6	6.2	13.4	19.1	22.6	24.0	22.9	22.2	17.4	13.3	6.6
	2%	DB	3.2	3.4	7.3	17.0	24.8	28.0	28.9	27.7	25.6	19.3	12.4	4.8
		MCWB	2.3	1.8	4.0	10.3	16.8	21.6	22.9	21.8	20.7	15.1	10.0	3.8
	5%	DB	1.7	1.9	5.3	14.1	22.2	26.0	27.3	26.3	23.5	16.7	10.0	3.2
		MCWB	0.9	0.8	2.8	9.3	15.4	19.9	21.8	21.0	19.1	13.4	8.0	2.1
	10%	DB	-0.2	0.7	3.6	11.7	20.1	24.2	25.8	24.8	21.6	14.5	7.9	1.9
		MCWB	-0.9	-0.2	1.7	7.3	14.3	18.7	20.8	20.1	18.0	11.9	6.2	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	4.2	7.8	15.1	20.8	24.3	25.6	24.5	23.1	18.3	13.8	7.0
		MCDB	5.6	5.0	9.6	19.4	25.7	28.6	29.4	28.1	26.8	21.4	15.2	7.6
	2%	WB	2.4	1.9	4.6	11.4	18.7	22.5	24.0	22.9	21.5	15.7	10.5	4.0
		MCDB	3.0	2.9	6.7	14.9	22.7	26.7	27.5	26.2	24.5	18.2	11.9	4.8
	5%	WB	0.8	1.0	3.1	9.6	17.0	21.1	22.7	21.9	20.1	14.0	8.0	2.4
		MCDB	1.5	1.7	4.7	13.3	20.6	24.8	26.0	25.0	22.9	16.2	9.5	3.0
	10%	WB	-1.0	-0.3	2.0	7.8	15.1	19.8	21.7	20.9	18.6	12.3	6.4	1.1
		MCDB	-0.3	0.6	3.3	11.1	18.9	23.0	25.0	23.9	21.1	14.2	7.8	1.7
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.3	8.2	8.5	9.8	9.6	9.4	9.6	9.7	7.7	6.3	6.7
		MCDBR	8.8	8.1	9.1	13.0	13.4	11.9	11.2	11.1	11.6	11.3	9.5	6.4
		MCWBR	8.8	7.5	6.7	8.2	7.3	6.8	6.5	6.5	7.3	7.6	8.1	6.3
	5% WB	MCDBR	8.7	7.9	8.3	11.8	11.2	10.5	10.0	9.9	10.7	10.0	8.5	6.5
		MCWBR	8.7	7.6	6.5	8.1	7.5	6.8	6.4	6.4	7.2	7.5	7.8	6.5
Clear-Sky Solar Irradiance	taub	0.259	0.285	0.317	0.351	0.378	0.416	0.419	0.408	0.375	0.335	0.308	0.279	
	taud	2.334	2.246	2.289	2.373	2.366	2.257	2.259	2.292	2.375	2.492	2.482	2.346	
	Ebn at Noon	850	888	909	906	888	852	844	838	837	828	788	776	
	Edh at Noon	74	99	111	113	118	133	131	122	103	79	66	66	
All-Sky Solar Radiation	RadAvg	1.42	2.33	3.61	4.47	5.13	5.62	5.56	4.88	3.83	2.23	1.33	0.97	
	RadStd	0.19	0.26	0.32	0.46	0.53	0.41	0.48	0.23	0.31	0.18	0.17	0.08	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page